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**Travelling tournament problem and its
NHL application**

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Abstract

This bachelor thesis focuses on the overview and application of the travelling tournament problem with a specific focus on its application to the National Hockey League. It provides a theoretical overview of the problem, its formal definition, variations and its complexity. The travelling tournament problem, which can be viewed as an extension of the better-known traveling salesman problem, aims to optimize sports tournament scheduling by minimizing travel distances. The thesis is structured in two main sections: the first discusses the theoretical aspects of the problem, including its computational complexity and various solving techniques. The second part applies these methodologies to the National Hockey League, proposing a model that reduces the total number of games and travel distances, potentially decreasing costs and player fatigue.

Keywords

National Hockey League, Tournament scheduling, Travelling tournament problem

JEL classification

C44, C61

Abstrakt

Tato bakalářská práce se zaměřuje na přehled a aplikaci Travelling tournament problem se specifickým zaměřením na jeho aplikaci na NHL. Poskytuje teoretický přehled problému, jeho formální definici, varianty a jeho složitost. Travelling tournament problem, který lze považovat za rozšíření známější úlohy obchodního cestujícího, má za cíl optimalizovat plánování sportovních turnajů minimalizováním procestovaných vzdáleností. Práce je strukturována do dvou hlavních částí: první se zabývá teoretickými aspekty problému, včetně jeho výpočetní složitosti a různých technik řešení. Druhá část aplikuje tyto metody na NHL a navrhuje model, který snižuje celkový počet zápasů a cestovní vzdálenosti, čímž potenciálně snižuje náklady a únavu hráčů.

Klíčová slova

NHL, Plánování turnajů, Travelling tournament problem

JEL klasifikace

C44, C61

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List of abbreviations

ANA Anaheim Ducks	NJD New Jersey Devils
ARI Arizona Coyotes	NSH Nashville Predators
BOS Boston Bruins	NYI New York Islanders
BUF Buffalo Sabres	NYR New York Rangers
CAR Carolina Hurricanes	OTT Ottawa Senators
CBJ Columbus Blue Jackets	PHI Philadelphia Flyers
CGY Calgary Flames	PIT Pittsburgh Penguins
CHI Chicago Blackhawks	SA Simulated annealing
COL Colorado Avalanche	SEA Seattle Kraken
DAL Dallas Stars	SJS San Jose Sharks
DET Detroit Red Wings	STL St Louis Blues
EDM Edmonton Oilers	TBL Tampa Bay Lightning
FLA Florida Panthers	TOR Toronto Maple Leafs
HAP Home-away pattern	TSP Travelling salesman problem
LAK Los Angels Kings	TTP Travelling tournament problem
MLB Major League Baseball	TTPPV Travelling tournament problem with predefined venues
MIN Minnesota Wild	VAN Vancouver Canucks
MTL Montreal Canadiens	VGK Vegas Golden Knights
NSH National Basketball Association	WIN Winnipeg Jets
NFL National Football League	WSH Washington Capitals
NHL National Hockey League	

Introduction

Sports have been an integral part of human civilization since ancient times, with events like the Olympic Games in Ancient Greece or gladiator fights and horse races in the Roman Colosseum. Over centuries sports and sports science has evolved, adapting to modern times with the aid of technology and research. As the sports industry grows, and more money is invested in it, so grow the expectations on both performance of the athletes and the revenues they generate. One such intersection of tradition, innovation and profit lies in the Traveling Tournament Problem (TTP), an extension of the Traveling Salesman Problem (TSP).

Put simply, the TTP focuses on creating an optimal schedule of a competition within the constraints of logistical efficiency. While the TSP aims to find the shortest possible route to visit a set of cities exactly once and return to the starting point, the TTP incorporates the proprieties of tournaments, where teams must adhere to schedules that not only minimize travel distances, but respect venue availability and satisfy the different team matchups. The main motivation for finding an optimal schedule can come from different points of view, whether we try to plan the most ecological routes with the lowest impact on the environment or try to have the players well rested to improve their performance on the field.

While the TTP is relevant across various sports leagues across the world, this thesis focuses specifically on showcasing how applying the TTP on the NHL could benefit the league. The primary objectives of this thesis are to present an overview of the TTP, model the problem using optimization software and utilizing data on the distances of NHL teams to highlight the current schedule inefficiencies. Ultimately, the thesis aims to illustrate how a TTP inspired approach could lead to a more equal and efficient schedule for the NHL.

The thesis is divided in two parts. The first focuses on the theory behind the TTP, its formal definition, an overview of the literature covering the problem, applications of the problem and its modelling. It also covers the complexity of the problem as well as computational complexity in general. A short section goes over scheduling methods used to generate feasible schedules and lastly programming in LINGO is introduced.

The second part of the thesis shifts its focus on the practical application of the TTP within a real sports league, specifically examining the National Hockey League (NHL). It explains the workings of the NHL schedule and the arguments of why shifting towards a TTP inspired schedule could benefit the league. Lastly, a detailed explanation of how to implement the TTP in LINGO is provided. This explanation goes over all the workings and parts of the model, and is followed by a short analysis of the results yielded by solving a smaller instance of the TTP. Solving this smaller instance should support the argumentation for changing the schedule.

1. The travelling tournament problem

This chapter focuses on introducing the Travelling tournament problem (TTP). It will go over the history of the TTP, its formal definition, its different variations and computing complexity. In the later parts, we will explain two methods used to generate schedules, look at literature covering the TTP and introduce programming in LINGO.

The primary objective of the travelling tournament problem is minimizing travelling distances while creating a feasible match schedule. That is why it is mainly used for scheduling leagues in large countries or geographical locations. In countries such as the United States the length of a road trip can differ up to hundreds of kilometers just by adding or subtracting one match. However we can still find applications for the TTP in smaller areas or countries, such as maximizing venue utilization or minimizing travel fatigue.

1.1 History

Sports scheduling problems have been mentioned in academic publications for the first time in 1974 by Eric Neil Gellig who worked with the concept of balanced tournament designs. The first publication on the topic of a scheduling problem was written by R.T. Campbell and D-S Chen in 1976. In their paper (Campbell and Chen, 1976), they analyzed scheduling the matches for a basketball league consisting of 10 teams in a double round robin tournament. Since then, there have been publications covering different aspects of the scheduling problems almost every year, with the numbers growing (Kendall et al., 2010) after the year 2001, when Kelly Easton and her colleagues formally described the Travelling tournament problem (Easton et al., 2001). They formulated the following definition:

Input: n , the number of teams; D an n by n integer distance matrix; L , U integer parameters.

Output: A double round robin tournament on the n teams such that

- The length of every home stand and road trip is between L and U inclusive and,
- The total distance traveled by the teams is minimized.

Easton also notes, that n must be even. For a better understanding of the formulation, we need to explain several key terms:

Round robin tournament: a tournament among n teams (n being even) where every team plays every other team. Such a tournament has $n/2$ games played and $n - 1$ slots in which the games are played.

Double round robin tournament: A double round robin tournament can be understood as two consecutive round robin tournaments where in the second half, the venues are switched, which means every team plays every other team twice, once home and once away. This kind

of tournament has $2(n - 1)$ slots and $n(n - 1)$ games played.

Home stand: A *home stand* consists of consecutive home games.

Road trip: A *road trip* consists of consecutive away games.

Length: The *length* of a road trip or home stand is the number of consecutive games played during said home stand or road trip.

With a better understanding of the TTP, we can look at a integer programming formulation of the standard TTP. The following formulation, which is a great and comprehensive mathematical formulation, can be found in **A polyhedral study of the Travelling Tournament Problem** by Marije Renske Siemann.

Data:

n : number of teams;

D : an n by n integer distance matrix;

U : the maximal length of a home stand or road trip;

Sets:

Slots $k \in \{1, \dots, 2(n - 1)\}$;

Teams $i, j, s, t \in \{1, \dots, n\}$;

Variables:

$$x_{k,i,j} = \begin{cases} 1, & \text{if team } i \text{ plays against team } j \text{ at home in slot } k \\ 0, & \text{otherwise} \end{cases} \quad \forall k, i, j$$

$$y_{i,s,t} = \begin{cases} 1, & \text{if team } i \text{ travels from } s \text{ to } t \\ 0, & \text{otherwise} \end{cases} \quad \forall i, s, \forall t \neq s$$

Objective function:

$$\min \sum_i \sum_s \sum_{t \neq s} d_{s,t} y_{i,s,t}$$

Constraints:

$$\sum_{j \neq i} (x_{k,i,j} + x_{k,j,i}) = 1 \quad \forall k, i \quad (1.1)$$

$$\sum_k x_{k,i,j} = 1 \quad \forall i, \forall j \neq i \quad (1.2)$$

$$x_{k,i,j} + x_{k+1,j,i} \leq 1 \quad k = 1, \dots, 2n - 3, \forall i, \forall j \neq i \quad (1.3)$$

$$\sum_{l=0}^U \sum_{j \neq i} x_{k+l,i,j} \leq U \quad k = 1, \dots, 2(n - 1) - U, \forall i \quad (1.4)$$

$$\sum_{l=0}^U \sum_{i \neq j} x_{k+l,i,j} \leq U \quad k = 1, \dots, 2(n-1) - U, \forall j \quad (1.5)$$

$$y_{i,s,t} \geq x_{k,s,i} + x_{k+1,t,i} - 1 \quad k = 1, \dots, 2n-3, \forall i, \forall s \neq i, \forall t \neq s, i \quad (1.6)$$

$$y_{i,i,t} \geq \sum_{j \neq i} x_{k,i,j} + x_{k+1,t,i} - 1 \quad k = 1, \dots, 2n-3, \forall i, \forall t \neq i \quad (1.7)$$

$$y_{i,t,i} \geq x_{k,t,i} + \sum_{j \neq i} x_{k+1,i,j} - 1 \quad k = 1, \dots, 2n-3, \forall i, \forall t \neq i \quad (1.8)$$

$$y_{i,i,t} \geq x_{1,t,i} \quad \forall i, \forall t \neq i \quad (1.9)$$

$$y_{i,t,i} \geq x_{2(n-1),t,i} \quad \forall i, \forall t \neq i \quad (1.10)$$

$$x_{k,i,j} \in \{0, 1\} \quad \forall k, i, j \quad (1.11)$$

$$y_{i,s,t} \in \{0, 1\} \quad \forall i, s, \forall t \neq s \quad (1.12)$$

Our objective function minimizes the total distance travelled by the teams. Constraint (1.1) makes every team play only once in each slot, at home or away. The next constraint (1.2) enforces that each team plays every other team twice, one game at home and one away. Constraint (1.3), the no-repeaters constraint, enforces that two teams do not play two consecutive games. This constraint is not always used, as schedules in many sports leagues all over the world do not adhere to it. Constraints (1.4) and (1.5) use U to enforce the maximum length of a home stand or road trip. Constraints (1.6), (1.7) and (1.8) couple the y and x variables. They manage trips from home to an opponents venue and vice versa. The constraints (1.9) and (1.10) are for the first and last matchups, because teams have to start and finish the tournament at home. These two constraints make sure a team travels from home to their first match and returns home if their last match is in an opponents venue. The last two constraints (1.11) and (1.12) make the variables x and y binary.

1.2 Variations

Ever since the problem was first proposed in 2001, researchers have come up with various models and variations with different constraints and objective functions. In this section we will go over the main variations and extensions of the TTP and propose a new variation.

The *mirrored travelling tournament problem* (Ribeiro and Urrutia, 2007) is a variant of the TTP with an additional constraint, which enforces that games played in round t are exactly the same as those played in round $t + (n-1)$ for $t = 1, \dots, n-1$ but with reversed venues. If we were to model a mirrored double round robin tournament, we would create two identical round robin tournaments, with the venues exactly reversed in the second round. Theoretically, this constraint should not affect the complexity of the problem, as we model only one round for the whole tournament and then merely flip it with each round. Schedules based on this variant are used mainly in sports leagues in Latin America.

The *travelling tournament problem with no repeaters* is a variation in which we have no teams i, j such that i plays at j and then j plays at i in the next slot (Easton et al., 2001). This kind of game planning is generally called a *home and home*, since we have two successive games with the same teams in which both teams will play one of the games at their respective home venue. Most sports leagues do not use this constraint when planning their schedule, as there have been many instances of home and home games in the past years in the NHL or in college football.

Travelling tournament problem with predefined venues (TTPPV) (Melo et al., 2009) is a single round robin variant of the TTP, in which the venue of each game to be played is known beforehand. This variant focuses only on creating a timetable which would minimize the objective function, since the home-away pattern (HAP) is predetermined. So the TTPPV aims to find a single round robin schedule compatible with a set of games that represents pairs of teams determined by the predefined HAP. The distance is minimized and no team plays more than three consecutive home or away games. In his paper, Melo proposed several integer programming formulations, one using a heuristic embedded in the mixed integer programming solver, other using the solver to enumerate solutions. He got similar computational results as Easton, his program easily solved for $n = 6$, while solving for $n \geq 8$ proved to be very hard.

Future variations

Even with numerous publications focusing on the TTP and sports scheduling, there has been a limited number of proposals suggesting new variations of the TTP. Given the imminent threat of global warming and the growing interest in sustainability and climate impact, an ecological variation of the TTP should be focused on cutting down CO₂ emissions. Such a variant could be mostly used in the north american sports leagues. This variant would need incorporating a new integer parameter, a corresponding constraint and a binary variable to determine the mode of transportation. The integer parameter would be the minimum length of a trip between two venues to use an airplane for travel. The constraint would enforce this minimum length with the binary variable. To illustrate, let's consider the NHL, where a team is visiting the New York area. Traditionally, the team would fly to New York, play against the two New York teams and the team from New Jersey, and then return home. However, for sustainability reasons, a more eco-friendly approach could involve, instead of using the plane, driving by bus to cover the relatively short distance of 144 kilometers to Philadelphia. The team would then playing an additional away game, and subsequently return to the NYC airport for the return flight. We know that the standard TTP is NP-complete, but there would be needed further research to see whether this variation would still remain NP-complete, given the additional parameter, constraint and variable.

1.3 Computational complexity

Computational complexity is the study of the resources needed, such as time and space, to solve computational problems. It also studies how these resources scale as the problems get bigger and more complex. The computing complexity of a problem can therefore be determined by the size of the problem at hand. The size can be seen as the number of variables and constraints the problem has. Put simply, a problem solving algorithm is effective if it is able to solve a problem in a reasonable amount of time. Based on the time needed to solve a problem, different problems are divided in the following four distinct categories; **P**, **NP**, **NP-complete** and **NP-hard**.

P problems are decision problems that can be solved in polynomial time. These are the simplest problems, that computers solve in our everyday lives. Finding the shortest path between two points on a map, sorting a list alphabetically or searching an item on a list.

NP problems are those, for solving which a non-deterministic polynomial algorithm exists. Finding the solution to these problems is not always easy, although the solutions are verifiable in polynomial time. The solving algorithms usually first generate a solution and then check whether it is the solution we are looking for. Examples of NP problems are the Hamiltonian path problem or the graph coloring problem.

NP-complete problems are problems which are both NP and NP-hard; we can view them as hard problems in the NP class. The NP class consists of all problems in NP for which it is possible to reduce any other NP problem in polynomial time. Examples of NP-complete problems are the Boolean satisfiability problem, Hamiltonian cycle or even Minesweeper (Kaye, 2000).

NP-hard problems are the most difficult problems to solve. This means that NP-hard problems are at least as hard as the hardest problem in the NP class. Although every problem in NP reduces to NP-hard, not all NP-hard problems are in NP. The most known examples of NP-hard problems are the Travelling salesman problem or the Halting problem.

Problem type	Verifiable in P time	Solvable in P time
P	Yes	Yes
NP	Yes	Yes or No*
NP-complete	Yes	Unknown
NP-hard	Yes or No**	Unknown***

Table 1.1: Increasing complexity of problems.

Source: (Stackoverflow, 2013)

*An NP problem that is also P is solvable in polynomial time.

** An NP-Hard problem that is also NP-Complete is verifiable in polynomial time.

*** NP-Complete problems (all of which form a subset of NP-hard) might be. The rest of NP hard is not.

Because of its many similarities to the TSP, the TTP has been considered to be NP-hard

since its proposal (Easton et al., 2001). We not only minimize the distance for one team (like we minimize for the traveller in the TSP) but for several teams and these minimized routes have to fit together to make a feasible schedule. The main two factors that affect the complexity of the TTP are the number of teams n and the parameter U , which gives us the maximal length of a road trip. While it is easy to understand and imagine how the computational complexity rises with the rising number of teams, understanding how the parameter U influences the computations can be trickier. The NP-completeness of the TTP has been proven only recently, but we will go over the main contributions towards this proof.

Bhattacharyya proved the NP-completeness of a reduced variant of the TTP where he got rid of the parameter U and the two constraints associated with it, that determine the maximum length of a home stand or road trip (Bhattacharyya, 2009). Even with this parameter and the two constraints removed, has Bhattacharyya shown how robust and computationally challenging the TTP is, being NP-complete even in this reduced version.

A year later, Thielen and Westphal proved the NP-completeness of the TTP when the upper bound for the parameter U is equal to 3 (Thielen and Westphal, 2011). They used this specific value due to it being the most common upper bound in most TTP models. They show NP-completeness of the TTP(3) by using a reduction from the boolean satisfiability problem (3-SAT).

Even if it is not yet peer reviewed, in 2021 Diptendu Chatterjee proved the NP-completeness of TTP(k), for any fixed $k > 3$ (Chatterjee, 2021). Chatterjee used a reduction from the k -SAT problem, but used different graph construction and trip structures than those used by Thielen and Westphal. This proof concludes the discussion about the complexity of the standard variation of the TTP.

1.4 Scheduling methods

There are many algorithms for generating tournament schedules. In this part of this thesis we will go over some well known scheduling methods, while actual schedules will be generated in the practical part of the thesis.

Kirkman tournament

One of the best known scheduling methods is named after Thomas Penyngton Kirkman. Kirkman discovered this method in 1846, even though he did not use it for scheduling tournaments, but rather as a solution for a mathematical problem of constructing triple systems of order n where $n = 1$ or 3 (Kirkman, 1847).

To construct a kirkman tournament, we create an empty array of $n - 1$ columns (n being the number of teams) and an unknown number of rows. The columns will represent our tourna-

ment slots. We fill the slots with matches ordered lexicographically, from left to right and from top to bottom. Each column represents a slot. To illustrate, we will create a schedule with a kirkman tournament with 4 teams.

Our array will have three columns and we start by placing the matches $\{1,2\}$, $\{1,3\}$ and $\{1,4\}$ in the first row. Next we want to schedule the match $\{2,3\}$. We cannot place it in the first or second column, as in the first slot, team 2 is already playing and in the second slot, team 3 is already playing. Therefore we place the match $\{2,3\}$ in the third column and move onto the next row. We place the match $\{2,4\}$ in the second column (again, team 2 is playing in the first slot) and then schedule the final match, between teams 3 and 4. We skip the third column again, as both team 3 and 4 play in the third slot. We place the game $\{3,4\}$ in the first column of the next row and we are finished.

Slot 1	Slot 2	Slot 3
$\{1, 2\}$	$\{1, 3\}$	$\{1, 4\}$
		$\{2, 3\}$
	$\{2, 4\}$	
$\{3, 4\}$		

Table 1.2: Kirkman tournament schedule for 4 teams.

Source: Own

In (Anderson, 1997) it was proved, that using this method with an even number of teams we can always get a feasible schedule for a single round-robin tournament. Creating a kirkman tournament by hand for 4 or 6 teams can be quick, creating such a schedule for more than 8 teams can be tricky. For a more clean image of the schedule we get rid of the empty spaces and move the matches up.

Slot 1	Slot 2	Slot 3
$\{1, 2\}$	$\{1, 3\}$	$\{1, 4\}$
$\{3, 4\}$	$\{2, 4\}$	$\{2, 3\}$

Table 1.3: Cleaner kirkman tournament schedule for 4 teams.

Source: Own

Circle method

The circle method is a simple and intuitive way of pairing teams to create a schedule. We place a team in the middle of a circle and place the rest of the teams on the outline of the circle. We pair the team in the middle with a team on the circle and then pair the neighbours of the outside team. Then, the neighbours of these neighbours play each other and so on, until we paired all the teams on the circle. This gives us the first round. After that, we merely rotate the circle to generate another round. We repeat rotations until the pairings are in their original position, at which point every team has played every other team. In

figure 1.1 below, we can see the first three rounds of an eight team tournament, where dots represent respective teams and the lines connecting them create matches. A rather elegant explanation of this method can be found in Scheduling a tournament (Froncek, 2010).

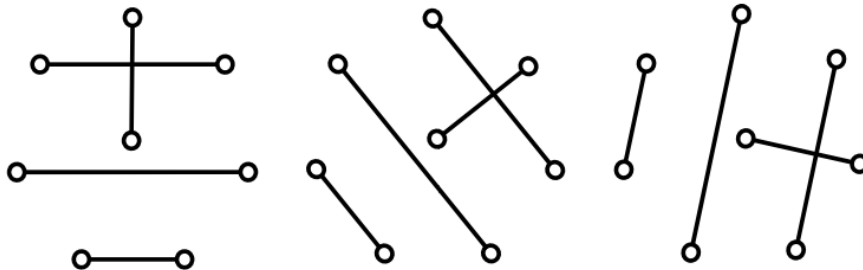


Figure 1.1: Circle method

Source: Scheduling a tournament, Dalibor Froncek

1.5 Literature review

The initial publications on sports scheduling surfaced in the 1970s. Since the proposal of the TTP in 2001, several papers and publications have addressed various aspects of this problem. It should also be noted, that doctor David Van Bulck with his research team at the Ghent University have created an online website (Bulck, 2016) covering sports scheduling and timetabling problems.

The paper, in which was the TTP proposed (Easton et al., 2001), has not only set up the basic theoretical model, but also proposed two approaches towards solving the problem. Easton explains that the TTP is a problem where we are faced both with the issues of feasibility and optimality. We need to find a feasible schedule when working with the home and away games while trying to minimize the total distance they travel. Easton also notes, that constraint programming excels at the former while integer programming does better at the latter, and proposes a constraint programming approach and a integer programming approach. While the constraint programming approach is useful in instances with a small n , for instances with more than 6 teams they would only approximate the optimal solutions. In their integer programming approach, Easton and her colleagues have added constraints to force every team to travel at least some specified lower bound which has improved the solving time. In this paper, they have shown that In general, size 4 instances are trivial, size 6 instances are difficult, and size 8 and larger instances are unsolved.

A year later, Easton et al. (2002), have published a paper with a combined integer programming and constraint programming approach. They used a branch and price algorithm, which is a column generation algorithm in which tours of each team are represented by the columns. Columns enter the basis to check whether the solution is optimal. With this ap-

proach, they have been able to solve instances up to 8 teams, which was an improvement on their paper from the year prior.

In 2003, Mieke Adriaen and her colleagues have compared and combined several meta-heuristic approaches into a hybrid, that performs rather well. In (Adriaen et al., 2003) they went over single neighbourhood experiments, which have similarities to random swapping and k-opt improvements when solving the TSP, swapping games, games days or inserting game days in free slots in the schedule. The second heuristic they used, is an agent based heuristic that was inspired by ant colony optimization techniques. Each game in the round-robin tournament gets assigned to an agent and is scheduled to be one of the games on the first day of the schedule. From there, each agent builds a timetable, choosing from candidate games on each step. The agent maintains information about the chosen games and the possible candidate games, and chooses games which have a higher desirability. This is influenced by the travelling distance, number of consecutive games and possible repeaters. Once all agents have created their timetables, the desirability of each game is updated and agents start over. The hybrid heuristic combines the constructive agent heuristic and the tabu search, which is an improvement heuristic. First the agent heuristic completes several iterations without the tabu search, after which the different swapping techniques are introduced. This hybrid has performed well and reached the optimal solution for small size instances of 4 and 6. Adriaen has found that combining the two approaches has yielded better results.

One of the better results for solving the TTP came in (Anagnostopoulos et al., 2006) from using a well known heuristic - simulated annealing. The main idea of simulated annealing is in accepting worse solutions at first in hopes of getting to a potentially better solution. It starts with a candidate solution - a randomly generated feasible schedule. Then it applies different improvements, swapping games, teams, rounds etc. Once a transformation of the solution is complete, the algorithm checks the total cost of the solution. If the cost is less than the candidate solution, it is accepted. If the cost is more, it is accepted or rejected with a given probability. This probability is determined by a parameter called temperature. At the beginning of SA, the temperature is assigned a higher value, which means a higher probability of accepting worse solutions. As the algorithm iterates, the temperature decreases over time, making it less likely to accept worse solutions. The idea is to explore the search space early in the process and not get stuck on a local minimum and later in the process converge to a locally optimal solution. The algorithm that Anagnostopoulos implemented uses the concept of reheating as well. After the algorithm iterates through the outermost loop, the temperature is increased to twice its value when the best solution is found. The reheating process stops after there is no improvement after a certain amount of reheats. At the time of the publication in 2006, Anagnostopoulos and his colleagues have improved all the best-known solutions on instances with at least 10 teams.

A great example of applying the TTP model on a sports league in the real world is a paper by Guillermo Durán (Durán et al., 2019). Durán and his colleagues modelled a new schedule for the top two professional basketball leagues in Argentina. Their research was funded by

the Argentine Basketball Club Association in hopes of improving the current schedules of the leagues. This association organizes the tournament of both of the basketball divisions in Argentina. The model created by Durán was implemented and improved the current system by saving over 40% of the total travel distance per away game. This decrease in distance not only reduces the fatigue of players but saves travel costs for all the teams as well.

Another publication worth mentioning is the **Scheduling in sports: An annotated bibliography** (Kendall et al., 2010). As the title suggests, this paper covers the main contributions and publications on different aspects of scheduling in sports. It goes over the different problems and definitions, follows up with methodologies and solving approaches and concludes with different applications on several sports.

1.6 LINGO programming

The practical part of this thesis is modelled in LINGO. This part will introduce LINGO and provide an explanation of why LINGO was chosen for modelling the TTP. Another obvious choice for a modelling software would be MPL for windows, but because the device on which this thesis was written does not support the windows operating system, LINGO was chosen instead.

LINGO is a programming language developed by the LINDO systems company (LINDO systems, 2024). This company provides tools and products for mathematical optimization. They have programs for linear programming, integer programming, quadratic programming, nonlinear programming, global and stochastic optimization. The name LINDO, represents what the company provides, Linear Interactive Discrete Optimizer. LINGO is a closed-source system, which means it does not typically support the integration of external solvers, because it has several different built-in solvers. One of the big advantages of using LINGO is, that the system reads your formulation of the problem and automatically chooses the appropriate solver. Although any optimization software is not cheap, LINDO systems provide free student trials for a limited time period. In this version, the user is not limited by the number of variables or constraints, but only by the capacity of the hardware.

The models in LINGO need to be written in a distinct programming language. Every model starts with the command `MODEL:` and ends with an `END`. Next the `SETS:` command is used to define the sets of the model, closed off with `ENDSETS`. The data of the model is in between `DATA:` and `ENDDATA`. Finally, the objective function and constraints are written using commands such as `MIN` or `MAX`, `@FOR`, `@SUM` and others (LINDO systems, 2020).

2. Travelling tournament problem for the National Hockey League

This part of the thesis focuses on solving a TTP while using data from a real sports league. Firstly we will introduce the National Hockey League (NHL), its scheduling as it is today and how and why applying the TTP on the NHL makes sense. Next we will go over the data from which a TTP model resurfaces. Lastly, using LINGO and its solver, we will solve a smaller instance of the problem.

2.1 The National Hockey League and its schedule

The NHL is the most famous hockey league in the world. It is based in North America, with 7 Canadian teams and 25 teams based in the United States. With a revenue north of 5 billion dollars in 2023 (Statista, 2024), it is expected to provide the best hockey experience in the world. The main source of this revenue are broadcasting rights, which is why a well prepared schedule is expected, so that the players are well rested and the games are played on prime-time to maximize viewership. The teams are located all around North America, not only in the metropolitan area on the east coast and Californian side on the west, but teams are based in several Canadian provinces, as well as two teams located in Florida. With such distances between the teams, even the difference of one game on a road trip can add or subtract thousands of kilometers when creating a schedule.

With the league having 32 teams, each playing 82 games in total, the schedule consists of 1312 total games that are played usually from the start of October until the beginning of April. All teams play half of their games at home and half on the road. The league is split into 2 conferences (Eastern and Western), each having 2 divisions (Atlantic and Metropolitan in the east, Central and Pacific in the west). It is clear, that with each team playing 82 games, the NHL schedule is not a classic TTP. Since the NHL spans across four time zones, the league has teams in the same conferences and divisions play each other more often. For example, a team from the Pacific division will play 32 games against the 16 teams from the Eastern conference. Half of those are at home and half away. Then the team will play 24 games against the 8 teams from the Central division, as those teams should be closer. The games are again split in half being played at home and half away. Of course, against some Central teams, the Pacific team will play two away games and one home game, while against other it will be the other way around. Lastly, the team will play 26 games against the 7 other teams from its division, playing 4 games (2 home and 2 away games) against 5 teams and 3 games against 2 teams. Below we can find a table showing the games a team from the Pacific division will play during a season. We can see, that half of the games are played at home and that more games are played in their own conference and division.

Pacific	H	A	Central	H	A	Metropolitan	H	A	Atlantic	H	A
ANA	2	2	ARI	2	1	CAR	1	1	BOS	1	1
CGY	1	2	CHI	1	2	CBJ	1	1	BUF	1	1
LAK	2	2	COL	1	2	NJD	1	1	DET	1	1
SEA	2	2	DAL	2	1	NYI	1	1	FLA	1	1
SJS	2	2	MIN	1	2	NYR	1	1	MTL	1	1
VAN	2	1	NSH	1	2	PHI	1	1	OTT	1	1
VGK	2	2	STL	2	1	PIT	1	1	TBL	1	1
EDM	X	X	WIN	2	1	WSH	1	1	TOR	1	1
Σ	13	13		12	12		8	8		8	8

Table 2.1: Example game count for the Edmonton Oilers
Source: Own

2.2 Scheduling discussion

In all parts of the NHL community, the news reporters, fans or even players, there are often discussions about the fairness of the schedule. There are two main arguments for changing the schedule from the current one into one that is more even and fair for all teams.

The first argument focuses on having an identical number of matches between all the teams. As mentioned before, the NHL is not scheduled as a classic double-round robin tournament. The teams play different opponents a different number of times based on which division and conference they are from. When looking at the map of the NHL, it is clear, that teams in the eastern conference travel overall much less than the teams in the western conference. The eastern teams are located much closer to each other while the teams from the western conference are much more spread out. Because teams are playing more matches against teams from their own division and conference, the schedule of the eastern conference teams is lighter from the travelling perspective; they travel less overall, whether in hours or kilometers. Looking at a analysis of the schedule for the 2022-2023 NHL season in (Barr, 2013) we can see, that the dispersion of the travel distances is quite large. Teams on the low end of the spectrum travelled around 34 000 miles overall, while the top 5 teams travelled over 46 000 miles overall. The main counterargument towards leaving different number of matches are rivalries between teams. When teams play each other more often, rivalries between them are born. These rivalries generate a better on ice product, which generates better revenues. With the way the NHL playoffs are set, teams from the same divisions often end up playing each other and their rivalries go on into the postseason.

In Figure 2.1, we can see the dispersion of NHL teams with an added imaginary line to divide the two conferences. North America is divided into four time zones and while the teams in the Western conference span across 3 of the four, all teams in the Eastern conference reside in the same time zone. This frequent changing of the time zone makes the travel of the western teams that much more exhausting.

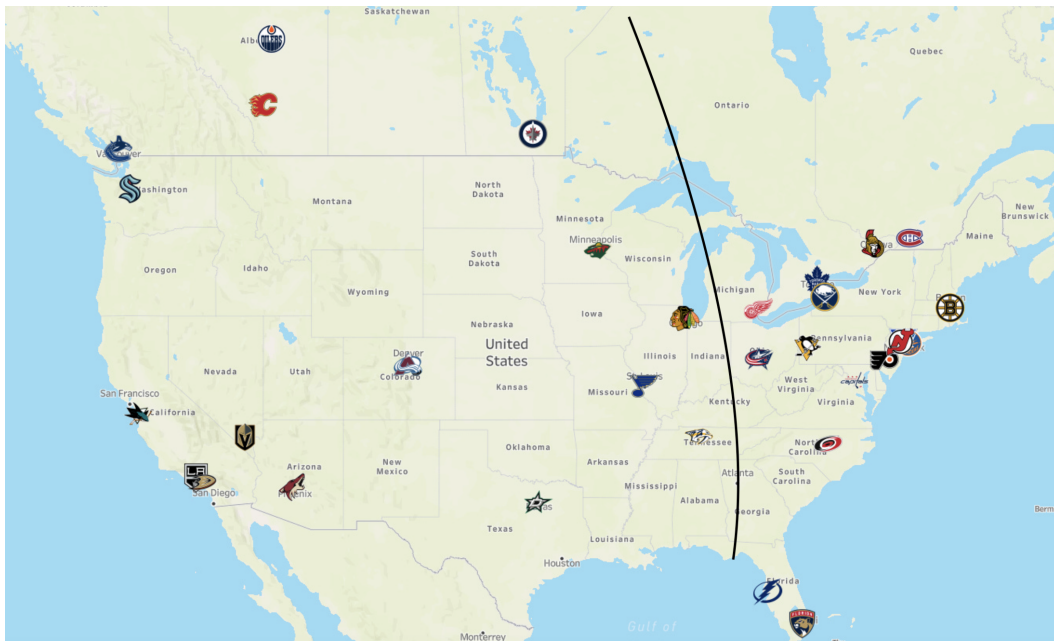


Figure 2.1: Map of NHL teams across North America
Source: (AZ Coyotes Insider, 2021)

The second argument focuses on lowering the total number of games in a season. The NHL is one of the big four leagues in North America (NHL, NBA, NFL, MLB) and while it has been around for more than a hundred years, it has the smallest total revenue out of the four. Many people argue that one of the main reasons why the NFL is the most viewed league of the 4, is less matches in a season. At first, this may seem counter-intuitive. But the less matches a season has, the more weight every win and loss has and the stakes are that much higher. Losing the first few matches drastically lowers the chances for a team to reach the playoffs. For comparison, the NBA has 30 teams that play 82 games each, MLB has 30 teams that each play 162 matches, but the NFL has 32 teams who play only 17 games each. As mentioned before, the NHL has an 82 game season and with so many games for each team, the pressure of winning becomes more relevant only towards the second half of the season.

The TTP seems to be a great response to both of these arguments. If the league were to implement a double round robin tournament, all teams would travel approximately an equal total distance and they would all play 62 games instead of 82. This would mean an equal travel load across the league and each game would have a greater impact on the final standings.

2.3 TTP for the NHL

Currently, the regular season of the NHL consists of 1312 games. A TTP model for the NHL would make up 62 matches for each team, lowering the total to 992 games. With the way the NHL schedule works right now, teams from different conferences already play each other twice, one home and one away game. The model would therefore take away only games between teams from the same conference.

Models of up to sixteen teams have been solved in polynomial time (Alatas and Bingol, 2019). Therefore the decision not to try to solve a model for the whole NHL, but only give a formal description was made. Because the TTP is NP-complete, no polynomial-time algorithm for the problem is known, which means an instance of 32 teams would take an impractically long time to solve. Because of this, rather than trying to solve the TTP for the whole league, models for instances of 4 teams from different divisions are made and the results between them are compared. Comparing the results between these models can show us how teams in different divisions travel much more.

All the models are based on the integer programming formulation of Marije Renske Siemann (Siemann, 2020). For the integer parameters, we use $U = 3$, which is the standard value used in most TTP models. For the distances in the distance matrix we use the distances between the individual arenas (deep13, 2024). The total distances are in kilometers.

2.4 Model with 32 teams

This part shows how a model with all the teams in the league would look like. This model has $62 \times 32 \times 31 = 61312$ $x_{k,i,j}$ variables, $62 \times 32 \times 31 = 61312$ $y_{i,s,t}$ variables and over 300 000 constraints. Running it on standard hardware would be not only impractical, but impossible as well. When this model is ran in the latest, 64 bit version of LINGO, an error occurred indicating not enough workspace in model generation.

Sets:

Slots $k \in \{1, \dots, 62\}$;

Teams $i, j, s, t \in \{1, \dots, 32\}$;

Variables:

$$x_{k,i,j} = \begin{cases} 1, & \text{if team } i \text{ plays against team } j \text{ at home in slot } k \\ 0, & \text{otherwise} \end{cases} \quad \forall k, i, j$$

$$y_{i,s,t} = \begin{cases} 1, & \text{if team } i \text{ travels from } s \text{ to } t \\ 0, & \text{otherwise} \end{cases} \quad \forall i, s, \forall t \neq s$$

Data:

$$n = 32$$

$$\mathbf{D} = \begin{bmatrix} 0 & 612 & 4233 & 3594 & 1955 & 3647 & 2852 & 1398 & \dots & 2519 \\ 612 & 0 & 3704 & 3072 & 1951 & 3062 & 2331 & 946 & & 2200 \\ 4233 & 3704 & 0 & 641 & 3363 & 983 & 1381 & 2842 & & 2163 \\ 3594 & 3072 & 641 & 0 & 2788 & 786 & 742 & 2200 & & 1597 \\ 1955 & 1951 & 3363 & 2788 & 0 & 3272 & 2235 & 1437 & & 1200 \\ 3647 & 3062 & 983 & 786 & 3272 & 0 & 1037 & 2342 & & 2158 \\ 2852 & 2331 & 1381 & 742 & 2235 & 1037 & 0 & 1462 & & 1144 \\ 1398 & 946 & 2842 & 2200 & 1437 & 2342 & 1462 & 0 & & 1285 \\ \vdots & & & & & & & & \ddots & \vdots \\ 2519 & 2200 & 2163 & 1597 & 1200 & 2158 & 1144 & 1285 & \dots & 0 \end{bmatrix}$$

$$U = 3$$

Objective function:

$$\min \sum_i \sum_s \sum_{t \neq s} d_{s,t} y_{i,s,t}$$

Constraints:

$$\sum_{j \neq i} (x_{k,i,j} + x_{k,j,i}) = 1 \quad \forall k, i \quad (2.1)$$

$$\sum_k x_{k,i,j} = 1 \quad \forall i, \forall j \neq i \quad (2.2)$$

$$x_{k,i,j} + x_{k+1,j,i} \leq 1 \quad k = 1, \dots, 61, \forall i, \forall j \neq i \quad (2.3)$$

$$\sum_{l=0}^3 \sum_{j \neq i} x_{k+l,i,j} \leq 3 \quad k = 1, \dots, 60, \forall i \quad (2.4)$$

$$\sum_{l=0}^3 \sum_{i \neq j} x_{k+l,i,j} \leq 3 \quad k = 1, \dots, 60, \forall j \quad (2.5)$$

$$y_{i,s,t} \geq x_{k,s,i} + x_{k+1,t,i} - 1 \quad k = 1, \dots, 61, \forall i, \forall s \neq i, \forall t \neq s, i \quad (2.6)$$

$$y_{i,i,t} \geq \sum_{j \neq i} x_{k,i,j} + x_{k+1,t,i} - 1 \quad k = 1, \dots, 61, \forall i, \forall t \neq i \quad (2.7)$$

$$y_{i,t,i} \geq x_{k,t,i} + \sum_{j \neq i} x_{k+1,i,j} - 1 \quad k = 1, \dots, 61, \forall i, \forall t \neq i \quad (2.8)$$

$$y_{i,i,t} \geq x_{1,t,i} \quad \forall i, \forall t \neq i \quad (2.9)$$

$$y_{i,t,i} \geq x_{62,t,i} \quad \forall i, \forall t \neq i \quad (2.10)$$

$$x_{k,i,j} \in \{0, 1\} \quad \forall k, i, j \quad (2.11)$$

$$y_{i,s,t} \in \{0, 1\} \quad \forall i, s, \forall t \neq s \quad (2.12)$$

2.5 Models with 4 teams

This section contains models for 4 teams. The difference between them lies purely in the distance matrices as the models are for teams from each of the divisions of the NHL. Solving the problem with smaller instances from each division provides a good comparison of the schedules for teams in different parts of North America. The 4 teams from each 8 team division were chosen alphabetically, that is first four teams from each division when in lexicographical order.

First, as in the previous section, the mathematical model is presented, while later we will analyze the code in LINGO for the model of one the divisions.

Data:

$$n = 4$$

$$\mathbf{A} = \begin{bmatrix} 0 & 641 & 984 & 1990 \\ 641 & 0 & 347 & 1863 \\ 984 & 347 & 0 & 1816 \\ 1990 & 1863 & 1816 & 0 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 0 & 2331 & 946 & 1439 \\ 2331 & 0 & 1462 & 1284 \\ 946 & 1462 & 0 & 1066 \\ 1439 & 1284 & 1066 & 0 \end{bmatrix},$$

$$\mathbf{M} = \begin{bmatrix} 0 & 753 & 790 & 768 \\ 753 & 0 & 37 & 15 \\ 790 & 37 & 0 & 22 \\ 768 & 15 & 22 & 0 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} 0 & 1955 & 2127 & 62 \\ 1955 & 0 & 172 & 1920 \\ 2127 & 172 & 0 & 2092 \\ 62 & 1920 & 2092 & 0 \end{bmatrix},$$

$$L = 1$$

$$U = 3$$

Sets:

Slots $k \in \{1, \dots, 6\}$;

Teams $i, j, s, t \in \{1, \dots, 4\}$;

Variables:

$$x_{k,i,j} = \begin{cases} 1, & \text{if team } i \text{ plays against team } j \text{ at home in slot } k \\ 0, & \text{otherwise} \end{cases} \quad \forall k, i, j$$

$$y_{i,s,t} = \begin{cases} 1, & \text{if team } i \text{ travels from } s \text{ to } t \\ 0, & \text{otherwise} \end{cases} \quad \forall i, s, \forall t \neq s$$

Objective function:

$$\min \sum_i \sum_s \sum_{t \neq s} d_{s,t} y_{i,s,t}$$

Constraints:

$$\sum_{j \neq i} (x_{k,i,j} + x_{k,j,i}) = 1 \quad \forall k, i \quad (2.13)$$

$$\sum_k x_{k,i,j} = 1 \quad \forall i, \forall j \neq i \quad (2.14)$$

$$x_{k,i,j} + x_{k+1,j,i} \leq 1 \quad k = 1, \dots, 5, \forall i, \forall j \neq i \quad (2.15)$$

$$\sum_{l=0}^3 \sum_{j \neq i} x_{k+l,i,j} \leq 3 \quad k = 1, \dots, 3, \forall i \quad (2.16)$$

$$\sum_{l=0}^3 \sum_{i \neq j} x_{k+l,i,j} \leq 3 \quad k = 1, \dots, 3, \forall j \quad (2.17)$$

$$y_{i,s,t} \geq x_{k,s,i} + x_{k+1,t,i} - 1 \quad k = 1, \dots, 5, \forall i, \forall s \neq i, \forall t \neq s, i \quad (2.18)$$

$$y_{i,i,t} \geq \sum_{j \neq i} x_{k,i,j} + x_{k+1,t,i} - 1 \quad k = 1, \dots, 5, \forall i, \forall t \neq i \quad (2.19)$$

$$y_{i,t,i} \geq x_{k,t,i} + \sum_{j \neq i} x_{k+1,i,j} - 1 \quad k = 1, \dots, 5, \forall i, \forall t \neq i \quad (2.20)$$

$$y_{i,i,t} \geq x_{1,t,i} \quad \forall i, \forall t \neq i \quad (2.21)$$

$$y_{i,t,i} \geq x_{62,t,i} \quad \forall i, \forall t \neq i \quad (2.22)$$

$$x_{k,i,j} \in \{0, 1\} \quad \forall k, i, j \quad (2.23)$$

$$y_{i,s,t} \in \{0, 1\} \quad \forall i, s, \forall t \neq s \quad (2.24)$$

At first glance we can see, that the teams from the western conference (Central and Pacific divisions), are spread out more than the ones from the eastern conference (Atlantic and Metropolitan divisions). We can therefore already assume a lower objective value of the models for the eastern teams.

It should be noted, that the constraints covering the maximum lengths of road trips and home stands, with the standard value of 3, are relevant only for models of instances with more than four teams. It is simply because in four team models, the longest road trip or home stand possible without any constraints, would still be three.

The current NHL schedule does not adhere to 3 constraints in our model. The NHL has repeaters - two subsequent games in which the same two teams play, only with reversed venues. It also does not have a constraint on the maximum length of road trips or home stands. For example, in the past 2023-2024 season, VAN had a 9 game home stand towards the end of the season, while DAL had a 4 game road trip.

LINGO model

As every model, this one, in Listing 2.1, starts with MODEL: and right after with the section SETS: to define all the sets in the model. The sets are collections of elements that are fundamental to the model, as they define the variables, parameters and indices for said variables and parameters. We are modelling for four teams which gives us $2(n - 1)$ slots. A match is represented by the variable x . It is played in a specific slot between two teams, which is reflected in how the variable is set up. The travel variable y has three indices as well, where team travels from one team to another. Finally we need to define that the distance matrix represents distances between two teams.

Listing 2.1: Sets of the model in LINGO

```
MODEL :
SETS :
teams /1..4/:team;
slots /1..6/:slot;
match(slots, teams, teams):x;
travel(teams, teams, teams):y;
distances(teams, teams):D;
ENDSETS
```

Next we define the data for the model in the DATA section. Here we are using the distances between the teams from the pacific division, namely teams from Anaheim, Calgary, Edmonton and Los Angeles.

Listing 2.2: Setting the data for the LINGO model

```
DATA :
D = 0 1955 2127 62
    1955 0 172 1920
    2127 172 0 2092
    62 1920 2092 0;
ENDDATA
```

Setting up the objective function may seem a bit tricky, but as we can see in Listing 2.3, the LINGO syntax makes it simple. The goal of our model is to minimize the travel distance, therefore we use the MIN command. With the sums we iterate through all the teams that play and the places they play at. Using NE, which indicates non-equality, we make sure a team does not travel from and to the same team in one match. The LINGO syntax makes using indices very simple, as we can see with the variables d and y .

Listing 2.3: Objective function minimizing the total travelled distance

```
MIN = @SUM(teams(i):
    @SUM(teams(s):
        @SUM(teams(t) | t #NE# s:
            D(s, t) * y(i, s, t))));
```

After setting up the objective function, we can start defining the respective constraints of our model. The first constraint ensures each team plays exactly once in each slot. The two for loops iterate over all slots and all teams. We sum the matches in each slot for the respective teams, whether the team is the home or the away team, making sure this sum equals one.

Listing 2.4: Each team plays once in each slot

```
@FOR(slots(k):
  @FOR(teams(i):
    @SUM(teams(j) | j #NE# i:
      x(k, i, j) + x(k, j, i))=1));
```

The next constraint we add ensures each team plays every other team exactly once at home and once away. We iterate through all home and away teams, and sum through all the slots.

Listing 2.5: Each team meets each other team exactly once at home and once away

```
@FOR(teams(i):
  @FOR(teams(j) | j #NE# i:
    @SUM(slots(k):
      x(k, i, j)) = 1));
```

The third constraint, in Listing 2.6, is the no-repeaters constraint. It ensures we do not schedule two consecutive matches, where first team i plays against j at home in one slot, with the match being repeated in the next slot with the venues reversed. First, we iterate through $2n - 3$ slots. This is because we do not need to check the last slot, whether a team has to play the same opponent in the subsequent slot, as there is no subsequent slot. We then iterate through all home and away teams. The binary variable $x_{k,i,j}$ equals 1 if team i plays against team j in slot k , and 0 otherwise. The sum $x_{k,i,j} + x_{k+1,j,i}$ counts the number of consecutive games between teams i and j across two rounds, k and $k + 1$. The constraint ensures that the sum of these variables is at most 1.

Listing 2.6: The no-repeaters constraint

```
@FOR(slots(k) | k #LE# 5:
  @FOR(teams(i):
    @FOR(teams(j) | j #NE# i:
      x(k, i, j) + x(k+1, j, i) <= 1));
```

The next two constraints are very similar, they enforce the maximum length of a home stand and a road trip, respectively. It is important to set the sums for each constraints to always reach up until the constant U which determines the maximum length of the consecutive matches played at home or away. As mentioned before, in the models for four teams are these constraints redundant, leaving them in or out of the model will not change the optimal solution.

Listing 2.7: Maximum length of a home stand

```
@FOR(slots(k) | k #LE# 3:
  @FOR(teams(i):
    @SUM(slots(l) | l #GE# 0 #AND# l #LE# 3:
      @SUM(teams(j) | j #NE# i:
        x(k+1, i, j))) <= 3));
```

Listing 2.8: Maximum length of a road trip

```
@FOR(slots(k) | k #LE# 3:
  @FOR(teams(j):
    @SUM(slots(l) | l #GE# 0 #AND# l #LE# 3:
      @SUM(teams(i) | i #NE# j:
        x(k+1, i, j))) <= 3));
```

The next constraint in Listing 2.9 is important to ensure the continuity in the travel schedule. It couples the y and x variables. It models the requirement that if a team travels from one opponent's venue to another in consecutive slots, the appropriate travel variables will reflect this. We iterate through all the needed slots and teams. It is important to reflect, with the @FOR conditions, that this condition applies only when teams travel from an opponent to a different opponent.

Listing 2.9: Travelling from an opponent to another opponent

```
@FOR(slots(k) | k #LE# 5:
  @FOR(teams(i):
    @FOR(teams(s) | s #NE# i:
      @FOR(teams(t) | t #NE# s #OR# i:
        y(i, s, t) >= x(k, s, i) + x(k+1, t, i) - 1))));
```

This constraint makes sure that the travel variables y correctly reflect the schedule variables x when teams travel from home to an opponent's venue. We need to iterate through the necessary slots and all teams.

Listing 2.10: Travel from home to an opponent

```
@FOR(slots(k) | k #LE# 5:
  @FOR(teams(i):
    @FOR(teams(t) | t #NE# i:
      y(i, i, t) >=
        @SUM(teams(j) | j #NE# i:
          x(k, i, j) + x(k+1, t, i) - 1))));
```

The next constraint is the opposite case of the previous constraint. It couples the y and x variables when teams travel from an opponent's venue back home. We subtract one from the sum to adjust the value to fit the binary constraint for y .

Listing 2.11: Travel from an opponent back home

```
@FOR(slots(k) | k #LE# 5:
  @FOR(teams(i):
    @FOR(teams(t) | t #NE# i:
      y(i,t,i) >=
        x(k,t,i) + @SUM(teams(j) | j #NE# i:
          x(k+1,i,j)) - 1)));
```

Because every team starts the tournament at home, half of the teams need to travel for the first match as well. The constraint in Listing 2.12 takes care of that. The constraint ensures that if team t plays against team i in the first slot of the tournament, then the variable indicating the travel of team i to team t must also be set to 1.

Listing 2.12: First match away

```
@FOR(teams(i):
  @FOR(teams(t) | t #NE# i:
    y(i,i,t) >= x(1,t,i)));
```

The last constraint is the opposite of the previous one. All teams must finish the tournament at home, so those, which play their last match away, need to travel back home. We therefore only check the match variables in the sixth (final) slot.

Listing 2.13: Last match away

```
@FOR(teams(i):
  @FOR(teams(t) | t #NE# i:
    y(i,t,i) >= x(6,t,i)));
```

Finally, in Listing 2.14 we set both variables to be binary with the @BIN command and indicate that our model is finished with the END command.

Listing 2.14: Types of variables

```
@FOR(match:@BIN(x));
@FOR(travel:@BIN(y));
END
```

2.6 Results

In this section we will go over the generated schedules for the respective divisions. For some, the differences may not seem significant, but that is mainly because of the way the teams for the shrunk divisions were chosen. For example, if we were to pick out the teams to find a significant difference between the results, we could choose the two New York teams, New Jersey and Philadelphia for the Metropolitan division and Arizona, Colorado, Dallas and Winnipeg for the Central division. That way one model would have an objective value below 1300 and the other over 23 000.

Atlantic division model

The model for the Atlantic division includes the teams from Boston, Buffalo, Detroit and Florida. The total amount of kilometers travelled by all teams is 19 960. The objective value is higher mainly because of the inclusion of the Florida team. LINGO found the solution in 9.26 seconds after roughly 142 000 iterations. The generated schedule can be found below.

	Match 1		Match 2	
Slot	Home Team	Away Team	Home Team	Away Team
1	DET	BUF	FLA	BOS
2	DET	FLA	BUF	BOS
3	DET	BOS	BUF	FLA
4	BOS	FLA	BUF	DET
5	BOS	DET	FLA	BUF
6	BOS	BUF	FLA	DET

Table 2.2: Generated schedule for the Atlantic division

Source: Own

As we can see, in both halves of the schedule there are teams that play all their home or away games consecutively. In this case, in the first three slots, Detroit plays all its home games and then goes on the road, while Boston has its schedule the other way around. Buffalo has its home games consecutively in the slots two to four. Florida is the only team, that starts with a home game, goes on the road for three games and then comes back home for the remaining two. The total distance travelled by respective teams is spread out rather equally. Both Florida and Detroit travel 4794 kilometers, while Boston and Buffalo travel 5184 and 5188 kilometers respectively.

Central division model

The Central division ended up with the highest value of the objective function of 21 593. If we go back and look at the map of the NHL franchises on page 20, it becomes clear why. While the total distance travelled by these teams is around 1600 kilometers more than those in the Atlantic division, their schedule is made much more difficult by the frequent crossing of the time zones. LINGO solves the model in approximately 7.7 seconds and it usually needs around 120 000 iterations to get to the optimal solution.

	Match 1		Match 2	
Slot	Home Team	Away Team	Home Team	Away Team
1	ARI	COL	DAL	CHI
2	ARI	DAL	COL	CHI
3	ARI	CHI	COL	DAL
4	CHI	DAL	COL	ARI
5	CHI	ARI	DAL	COL
6	CHI	COL	DAL	ARI

Table 2.3: Generated schedule for the Central division

Source: Own

Just as with the schedule for the Atlantic division, two teams have their first three games either at home or on the road. Two teams, Arizona and Dallas, travel 5131 kilometers in total. Chicago travels 5627 kilometers, while Colorado travels the most of all the teams, 5704 kilometers. When we look at the map of the NHL teams, we can see that Denver (the capital of Colorado, where the NHL team resides) is the most secluded out of all the teams in the league. While it is in the middle of the map, just like Dallas, it does not have any team in close proximity. This means that the team needs to have longer road trips, as road trips with only one match are way less efficient.

Metropolitan division model

This division, gets the easiest schedule from the travelling point of view. The objective value, 8366 kilometers, is the lowest out of all the models. Even if we modelled the schedules for the whole divisions, the Metropolitan division would have the lowest objective value. It is because all of the teams from this division are located much closer than others. The eight teams of this division are located in six neighbouring states. For comparison, the Central division has seven teams in different states and one team in Canada. The model was solved in around 8.7 seconds with all four teams travelling between 2084 and 2099 kilometers.

	Match 1		Match 2	
Slot	Home Team	Away Team	Home Team	Away Team
1	CAR	NYI	CBJ	NJD
2	CAR	NJD	CBJ	NYI
3	CAR	CBJ	NJD	NYI
4	NYI	CAR	NJD	CBJ
5	NYI	CBJ	NJD	CAR
6	NYI	NJD	CBJ	CAR

Table 2.4: Generated schedule for the Metropolitan division
Source: Own

Pacific division model

The model for the final division, the Pacific, is interesting mainly from a geographical standpoint. It includes two Canadian teams from the province of Alberta and two Californian teams situated merely 62 kilometers from each other. That way, we have two pairs of teams close to each other, which would make approximating the optimal solution by hand really easy. The objective value of this model is 17 124, with all four teams travelling exactly 4281 kilometers. The model was solved in just under 7 seconds.

	Match 1		Match 2	
Slot	Home Team	Away Team	Home Team	Away Team
1	EDM	CGY	ANA	LAK
2	EDM	LAK	CGY	ANA
3	EDM	ANA	CGY	LAK
4	LAK	ANA	CGY	EDM
5	LAK	EDM	ANA	CGY
6	LAK	CGY	ANA	EDM

Table 2.5: Generated schedule for the Pacific division
Source: Own

When we look at the generated schedule, we can see a very intuitive layout of the matches. In the first few slots, the Californian teams play their matches in Canada, while in the second half of the schedule the matches move south and the Canadian teams play their away games in California.

Key takeaways

The four models have many similarities as well as some differences. The main takeaway is that in all four models, three teams fully fill the maximum length of road trips or home stands. As mentioned above, the obvious outline is the Metropolitan division, which has a

much lower objective value than the other three divisions. It is important to note, that for these smaller instances, the expectation of a more balanced travel schedule was met. In the generated schedules, all teams travel almost the same distance. We can assume that if we were to solve a model for the whole league, the distances would be close for all the teams again. A schedule with a balanced travel load for all the teams is what the TTP for the whole league would provide.

It should be noted, that reduced costs hold no value for us, as they always just hold the value of the travel of a different potential match up.

2.7 Models with 6 and 8 teams

The LINGO model above, was adjusted for instances of both 6 and 8 teams. For the eight team model, the maximum length of road trips and home stands constraints were omitted. For the six team model, these constraints were kept, although U was set to 4. A longer solving time was expected, but solving both models had to be interrupted for practical reasons. The six team model solver was interrupted after 30 hours, while the eight team model solver ran for 75 hours before being interrupted. The distances for the model with six teams can be seen below in the table 2.6. This solution is not optimal, an example of a part that could see improvement is San Jose playing in Edmonton, coming back home and right after that travelling to Calgary. Switching these last two matches, would bring the objective value down, although the schedule of the other teams would need to be adjusted accordingly.

Team	Distance travelled
ANA	4694
CGY	5446
EDM	6197
LAK	5549
SEA	7723
SJS	6815
All teams	36 424

Table 2.6: 6 team Pacific division travel distances

Source: Own

The model for eight teams had an objective value of 79 623 kilometers. This result makes every team travel in average 10 000 kilometers in total. Again, the objective value is not optimal, but gives us a rough estimate of what it would approximately be. In this case, the distances are less spread apart, with no team travelling less than 8400 kilometers and no team travelling more than 12 200 kilometers. In the model, most home stands and road trips do not exceed the number 6. The full LINGO model can be found in appendix A. LINGO model.

Conclusion

This thesis provides a comprehensive overview and explanation of the Travelling tournament problem, specifically applied to the National Hockey League. The primary goal of this problem is to optimize the schedule of a sports tournament to minimize travel distance while adhering to various constraints. This thesis demonstrates how applying the TTP to the NHL could lead to reduction of the total travel distance, game count, and associated travel costs while balancing the travel schedule across the league.

We examine the main variants of the TTP, while proposing a novel one, which could focus on the ecological impact by reducing CO₂ emissions, reflecting an important integration of sustainability into sports scheduling. Additionally, the thesis explains the computational complexity of the TTP, classifying it as an NP-complete problem, and highlights the significant computational effort required to solve larger instances of the problem.

In the thesis, we look at the main arguments for applying the TTP to the NHL and the benefits and positives it could bring. We successfully apply integer programming and optimization software (LINGO) to model the NHL's scheduling, breaking down the theoretical problem into a practical solution. By modeling the NHL in smaller 4-team instances, we showcase the potential benefits of a more balanced and efficient schedule. Results from these models reveal differences in travel demands across divisions, indicating room for improvement in the current schedule.

The detailed explanation of the LINGO model provides a solid starting point for further modeling and analysis of different variants of the TTP in LINGO. As mentioned in chapter 1.5, the TTP has already been successfully applied on a real sports league, yielding promising results. Thus, it can be concluded that the TTP has potential for application not only on the National Hockey League, but many different sports leagues worldwide. Further research into sports timetabling and the TTP specifically should be in the interest of all the people in the sports community, including fans, team owners, broadcasters or just amateur sports science enthusiasts.

Bibliography

- ADRIAEN, Mieke, CUSTERS, Nele and BERGHE, G Vanden, 2003. An agent based meta-heuristic for the traveling tournament problem. In: *Workshop on real-life applications of Metaheuristics, Antwerpen*. Citeseer.
- ALATAS, Bilal and BINGOL, Harun, 2019. A physics based novel approach for travelling tournament problem: Optics inspired optimization. *Information Technology and Control*. Vol. 48, no. 3, pp. 373–388.
- ANAGNOSTOPOULOS, Aris, MICHEL, Laurent, HENTENRYCK, P Van and VERGADOS, Yannis, 2006. A simulated annealing approach to the traveling tournament problem. *Journal of Scheduling*. Vol. 9, pp. 177–193.
- ANDERSON, Ian, 1997. *Combinatorial designs and tournaments*. Oxford University Press. No. 6.
- AZ COYOTES INSIDER, LLC, 2021. *NHL divisional alignment: Which format is best?* [online]. [visited on 2024-04-27]. Available from: <https://azcoyotesinsider.substack.com/p/nhl-divisional-alignment-which-format>.
- BARR, John, 2013. *NHL schedule* [online]. [visited on 2024-05-03]. Available from: <https://soundofhockey.com/2022/10/11/2022-23-nhl-schedule-breakdown/>.
- BHATTACHARYYA, Rishiraj, 2009. A note on complexity of traveling tournament problem. *Optimization Online*. Vol. 2480.
- BULCK, David van, 2016. *RobinX: an XML-driven classification for round-robin sports timetabling* [online]. [visited on 2024-04-27]. Available from: <https://robinxval.ugent.be/RobinX/>.
- CAMPBELL, Robert Thomas and CHEN, DS, 1976. A minimum distance basketball scheduling problem. *Management science in sports*. Vol. 4, pp. 15–26.
- DEEP13, 2024. *NHL team distances* [online]. [visited on 2024-04-27]. Available from: https://www.deep13.xyz/nhldistance#pacific_division.
- DURÁN, Guillermo, DURÁN, Santiago, MARENCO, Javier, MASCIALINO, Federico and REY, Pablo A, 2019. Scheduling Argentina’s professional basketball leagues: A variation on the Travelling Tournament Problem. *European Journal of Operational Research*. Vol. 275, no. 3, pp. 1126–1138.
- EASTON, Kelly, NEMHAUSER, George and TRICK, Michael, 2001. The traveling tournament problem description and benchmarks. In: *Principles and Practice of Constraint Programming—CP 2001: 7th International Conference, CP 2001 Paphos, Cyprus, November 26–December 1, 2001 Proceedings 7*. Springer, pp. 580–584.
- EASTON, Kelly, NEMHAUSER, George and TRICK, Michael, 2002. Solving the travelling tournament problem: A combined integer programming and constraint programming approach. In: *International conference on the practice and theory of automated timetabling*. Springer, pp. 100–109.

- FRONCEK, Dalibor, 2010. Scheduling a tournament. *Dolciani mathematical expositions*. Vol. 43, pp. 203–216.
- CHATTERJEE, Diptendu, 2021. Complexity of traveling tournament problem with trip length more than three. *arXiv preprint arXiv:2110.02300*.
- KAYE, Richard, 2000. Minesweeper is NP-complete. *Mathematical Intelligencer*. Vol. 22, no. 2, pp. 9–15.
- KENDALL, Graham, KNUST, Sigrid, RIBEIRO, Celso C and URRUTIA, Sebastián, 2010. Scheduling in sports: An annotated bibliography. *Computers & Operations Research*. Vol. 37, no. 1, pp. 1–19.
- KIRKMAN, Thomas P, 1847. On a problem in combinations. *Cambridge and Dublin Mathematical Journal*. Vol. 2, pp. 191–204.
- LINDO SYSTEMS, inc., 2020. *LINGO the modelling language and optimizer*. LINDO SYSTEMS inc.
- LINDO SYSTEMS, inc., 2024. *About Lindo Systems* [online]. [visited on 2024-04-27]. Available from: <https://www.lindo.com/index.php/company>.
- MELO, Rafael A, URRUTIA, Sebastián and RIBEIRO, Celso C, 2009. The traveling tournament problem with predefined venues. *Journal of Scheduling*. Vol. 12, no. 6, pp. 607–622.
- RIBEIRO, Celso C and URRUTIA, Sebastián, 2007. Heuristics for the mirrored traveling tournament problem. *European Journal of Operational Research*. Vol. 179, no. 3, pp. 775–787.
- SIEMANN, Marije Renske, 2020. *A polyhedral study of the Travelling Tournament Problem*. MA thesis. University of Twente.
- STACKOVERFLOW, 2013. *Problem complexity* [online]. [visited on 2024-04-20]. Available from: <https://stackoverflow.com/a/19510170>.
- STATISTA, 2024. *NHL total revenue* [online]. [visited on 2024-04-27]. Available from: <https://www.statista.com/statistics/193468/total-league-revenue-of-the-nhl-since-2006/>.
- THIELEN, Clemens and WESTPHAL, Stephan, 2011. Complexity of the traveling tournament problem. *Theoretical Computer Science*. Vol. 412, no. 4-5, pp. 345–351.

Appendices

A. LINGO model

```
MODEL:
SETS:
teams /1..8/:team;
slots /1..14/:slot;
match(slots , teams , teams):x;
travel(teams , teams , teams):y;
distances(teams , teams):distance;
ENDSETS

DATA:
!Pacific division;
distance = 0 1955 2127 62 1563 478 1755 582
          1955 0 172 1920 710 1646 673 1441
          2127 172 0 2092 835 1813 764 1611
          62 1920 2092 0 1548 487 1740 530
          1563 710 835 1548 0 1145 192 1208
          478 1646 1813 487 1145 0 1331 597
          1755 673 764 1740 192 1331 0 1394
          582 1441 1611 530 1208 597 1394 0;
ENDDATA

MIN = @SUM(teams(i): @SUM(teams(s): @SUM(teams(t) | t #NE# s:
distance(s, t) * y(i, s, t))));

! Each team plays once in each slot;
@FOR(slots(k):@FOR(teams(i):@SUM(teams(j) | j #NE# i:
x(k, i, j) + x(k, j, i))=1));

! Each team meets each other team exactly once at home and once away;
@FOR(teams(i):@FOR(teams(j) | j #NE# i:@SUM(slots(k): x(k, i, j)) = 1));

! No repeaters constraint;
@FOR(slots(k) | k #LE# 13:@FOR(teams(i):@FOR(teams(j) | j #NE# i:
x(k, i, j) + x(k+1, j, i)<=1));

! Maximum home stand length;
@FOR(slots(k) | k #LE# 11:@FOR(teams(i):
@SUM(slots(l) | l #GE# 0 #AND# l #LE# 3:@SUM(teams(j) | j #NE# i:
x(k+l, i, j))) <= 3));

! Maximum road trip length;
@FOR(slots(k) | k #LE# 11:@FOR(teams(j):
@SUM(slots(l) | l #GE# 0 #AND# l #LE# 3:@SUM(teams(i) | i #NE# j:
x(k+l, i, j))) <= 3));
```

```
! Travel from opponent to another opponent;
@FOR(slots(k) | k #LE# 13:@FOR(teams(i):@FOR(teams(s) | s #NE# i:
@FOR(teams(t) | t #NE# s #OR# i:
y(i,s,t) >= x(k,s,i) + x(k+1,t,i) - 1)))));

! Travel from home to an opponent;
@FOR(slots(k) | k #LE# 13:@FOR(teams(i):@FOR(teams(t) | t #NE# i:
y(i,i,t) >= @SUM(teams(j) | j #NE# i:
x(k,i,j)) + x(k+1,t,i) - 1)))));

! Travel from an opponent back home;
@FOR(slots(k) | k #LE# 13:@FOR(teams(i):@FOR(teams(t) | t #NE# i:
y(i,t,i) >= x(k,t,i) + @SUM(teams(j) | j #NE# i:
x(k+1,i,j)) - 1)))));

! First game away;
@FOR(teams(i):@FOR(teams(t) | t #NE# i: y(i,i,t) >= x(1,t,i)));

! Last game away;
@FOR(teams(i):@FOR(teams(t) | t #NE# i: y(i,t,i) >= x(14,t,i)));

@FOR(match:@BIN(x));
@FOR(travel:@BIN(y));

END
```